class-3

3) Determine the phase relationship between the sinusoidal waveforms of each of the following sets.

$-\sin\omega t = \sin(\omega t \pm 180^\circ)$

$-\cos\omega t = \cos(\omega t \pm 180^\circ)$

$\mp \sin\omega t = \cos(\omega t \pm 90^\circ)$

$\pm \cos\omega t = \sin(\omega t \pm 90^\circ)$

c. $i = 2\cos(\omega t + 10^\circ)$
$v = 2\sin(\omega t - 10^\circ)$

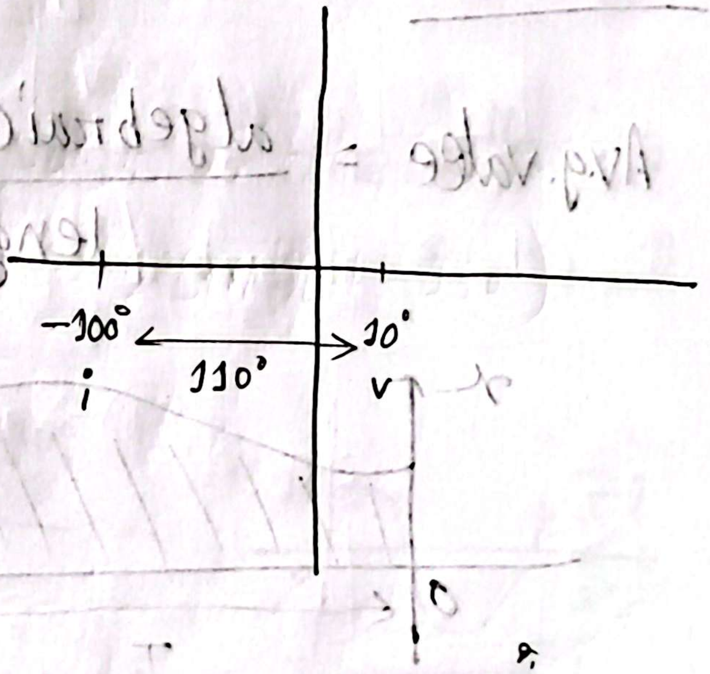
e. $i = -2\cos(\omega t - 60^\circ)$
$v = 3\sin(\omega t - 150^\circ)$

Solve

c) $i = 2\cos(\omega t + 10^\circ)$

$= 2\sin(\omega t + 10 + 90^\circ)$

$= 2\sin(\omega t + 100^\circ)$



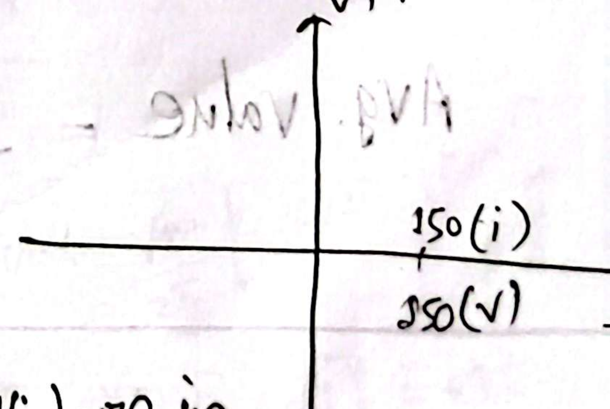
$v = 2\sin(\omega t - 10^\circ)$

$\begin{cases} i \text{ leads } v \text{ by } 110^\circ \\ v \text{ lags } i \text{ by } 110^\circ \end{cases}$

e) $i = -2\cos(\omega t - 60^\circ)$

$= 2\sin(\omega t - 60^\circ - 90^\circ)$

$= 2\sin(\omega t - 150^\circ)$

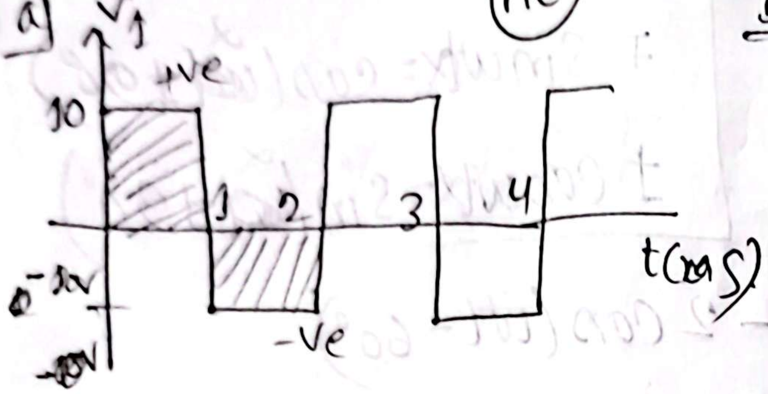


$v = 3\sin(\omega t - 150^\circ)$

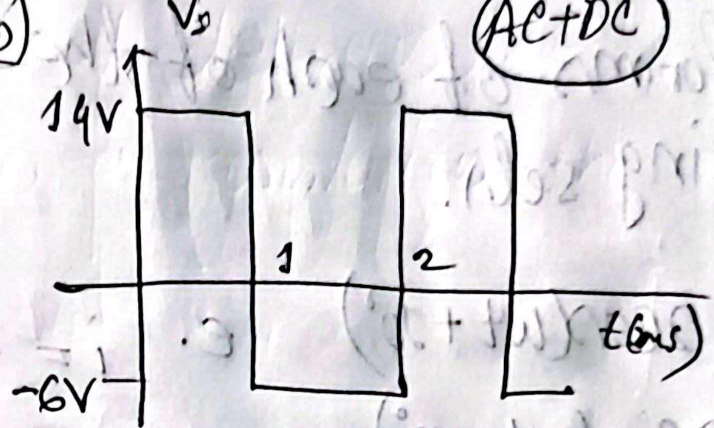
(v) and (i) are in Phase. Ans

*] Determine the average value of the waveforms.

a) (AC)

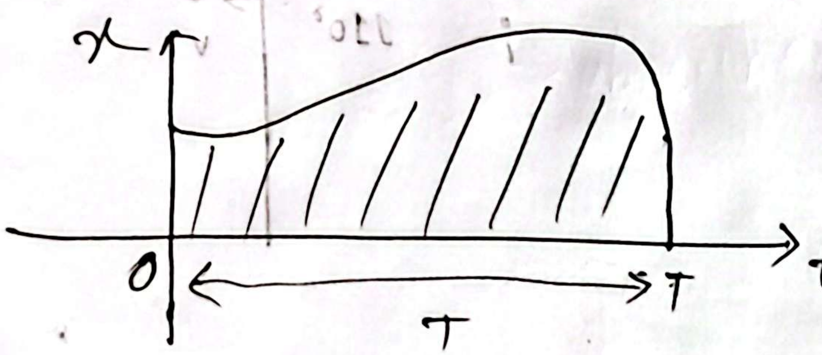


b) (AC+DC)



Avg. value

$$\text{Avg. valee} = \frac{\text{algebraic sub of areas}}{\text{length of curve.}}$$



$$\text{Avg. value} = \frac{\int_0^T x\,dt}{T}$$

Ans

a) $\text{Avg value} = \dfrac{(10\times1)-(10\times1)}{2}$

$= 0\,V$ Ans

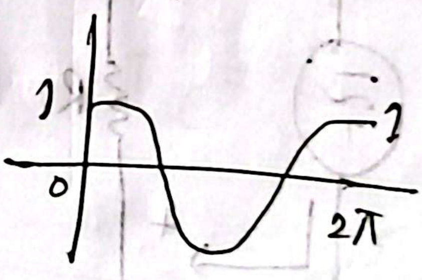
b) $\text{Avg value} = \dfrac{(14\times1)-(6\times1)}{2}$

$= 4\,V$. Ans

Q) Determine the average value of the following signal. $v(t) = V_m \sin\alpha$

Ans

$$V_{avg} = \frac{\int_0^{2\pi} v(\alpha)\, d\alpha}{2\pi}$$



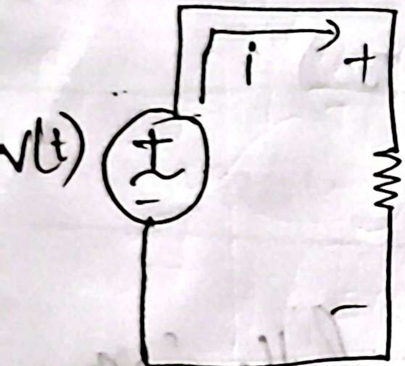
$$= \frac{\int_0^{2\pi} V_m \sin\alpha \, d\alpha}{2\pi}$$

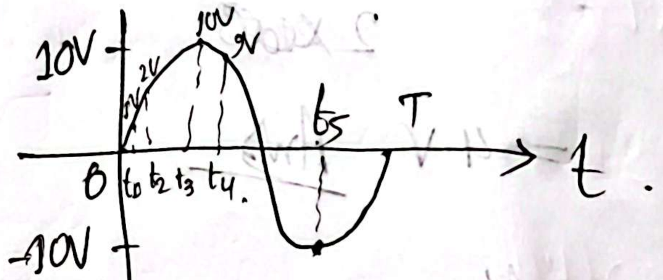
$$= \frac{V_m}{2\pi}\int_0^{2\pi} \sin\alpha \, d\alpha \quad = \frac{V_m}{2\pi}\left[-\cos 2\pi + \cos 0\right]$$

$$= \frac{V_m}{2\pi}\left[-\cos\alpha\right]_0^{2\pi} = \frac{V_m}{2\pi}\left[-1+1\right] = 0$$

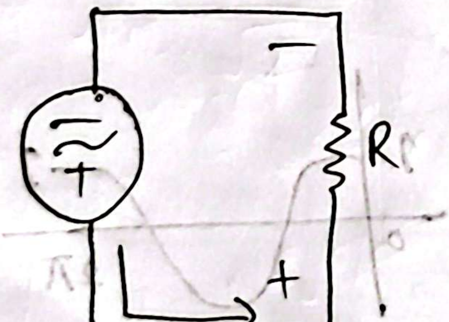
Effective value / RMS (Root mean Square) value

The effective value of a periodic current is the DC current that delivers the same avg. power to a resistor as the periodic current

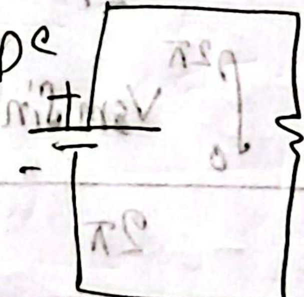


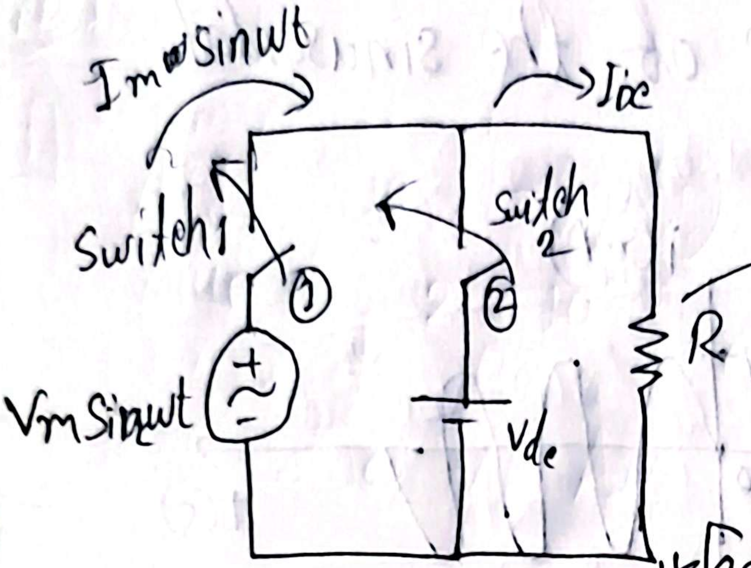


P = (+) → absorbed



P(+) → absorbed.





→ temperature (T_1)

$i_{ac} = I_m \sin\omega t$ [Sinusoidal current]

#. [generic AC waveform in next page]

$P_{ac(av)} = (i_{ac})^2 R$ ①

$= (I_m \sin\omega t)^2 R$

$= I_m^2 R \cdot \sin^2\omega t$

$= I_m^2 R \cdot \frac{1}{2} 2\sin^2\omega t$

$= \frac{I_m^2 R}{2}(1 - \cos 2\omega t)$

a.v = average value

$= \frac{I_m^2 R}{2} - \underbrace{\frac{I_m^2 R}{2} \boxed{\cos 2\omega t}}_{0}$

t_1, t_2, t_3
$\downarrow \; \downarrow \; \downarrow$
$P_1 \; P_2 \; P_3$

$= \frac{I_m^2 R}{2}$

$P_{ac(av)}$

$= \frac{1}{T}\left[\int_0^T \frac{I_m^2 R}{2} dt + \int_0^T \frac{I_m^2 R}{2} \underbrace{\cos 2\omega t}_{} dt\right]$

Average value of the cosine wave is zero

$= \frac{1}{T} \cdot \frac{I_m^2 R}{2} \int_0^T dt$

$= \frac{1}{T} \cdot \frac{I_m^2 R}{2} \cdot T$

$= \frac{I_m^2 R}{2}$

$P_{DC} = I_{dc}^2 R$

$P_{ac(av)} = P_{DC}$ ②

$\Rightarrow \frac{I_m^2 R}{2} = I_{dc}^2 R$ ③ → effective value / rms value

$\Rightarrow \boxed{I_{dc} = \frac{I_m}{\sqrt{2}}}$

$I_{RMS} = I_{dc} = \frac{I_m}{\sqrt{2}}$

$= 0.707 I_m$

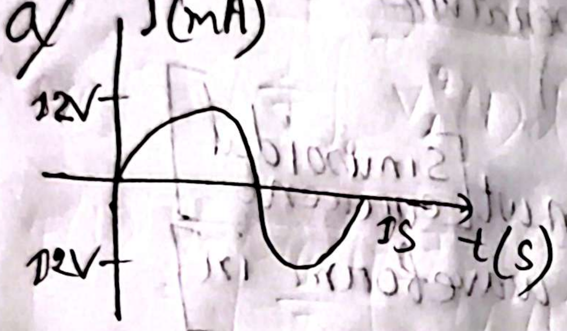
I_m = Current peak Amplitude.

$I_{eff} = \frac{I_m}{\sqrt{2}} = 0.707 I_m$

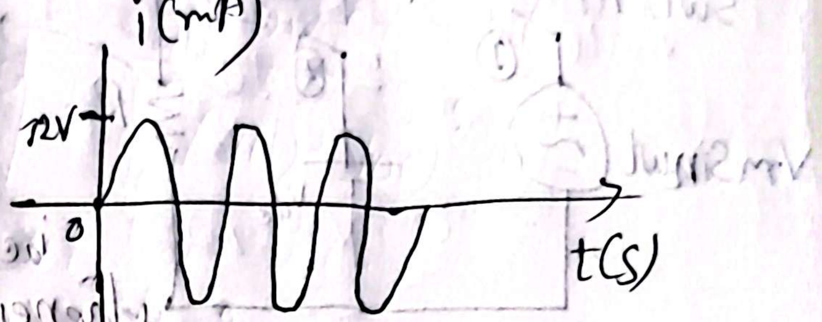
↳ effective current

Find the rms value of the sinusoidal waveform.

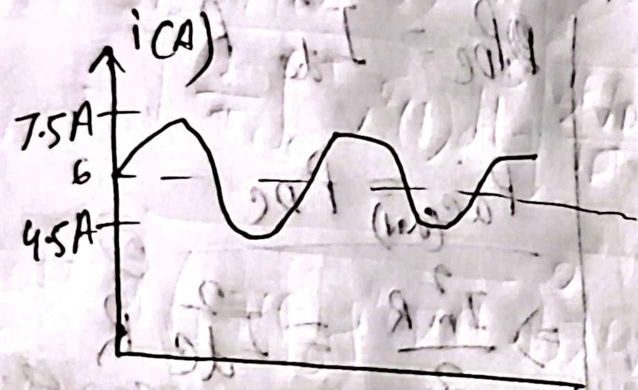
a/



b/



c/



~ AC+DC (6A)

Ans

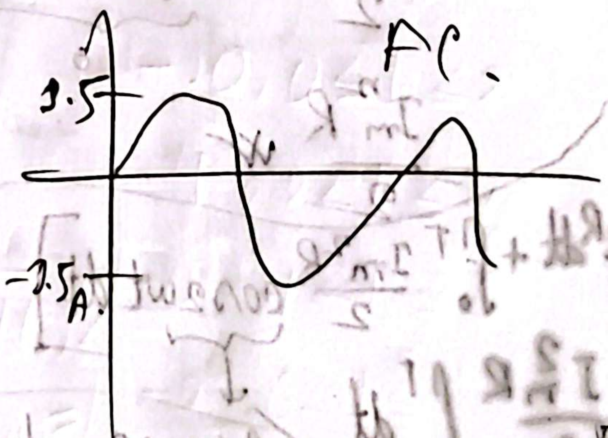
a/ $I_{Rms} = \frac{12}{\sqrt{2}}$

$= 6\sqrt{2}\ mA$

b/ $I_{Rms} = \frac{12}{\sqrt{2}}$

$= 6\sqrt{2}\ mA$

c/ $I_{Rms} = \sqrt{I^2_{ac\,(rms)} + I^2_{dc\,(rms)}}$



$= \sqrt{\left(\frac{1.5}{\sqrt{2}}\right)^2 + (6)^2}$

$=$